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REPORT
FINAL ASSIGNMENT
MACHINE VISION

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CHAPTER I: ASSIGNMENT OVERVIEW

This report presents the methodology and experimental results of the project entitled "Development of a Heatmap for Analyzing Badminton Players' Activity Areas." The project was researched and developed by a two-member team consisting of Tran Minh Nhat and Trinh Minh Tri.

To generate a heatmap that accurately reflects players' movement patterns, the primary challenge is to determine the player's position precisely within the image. However, since the input data are typically captured by surveillance cameras installed at oblique viewing angles, perspective distortion causes significant geometric discrepancies compared with a top-down (bird's-eye) view. To address this issue, the proposed approach extracts the relative coordinates of the player's feet—which represent the contact points with the court surface—using the court corners as reference points. Based on these reference coordinates, geometric transformation techniques are applied to map the image coordinates onto the actual coordinates of the badminton court. The resulting spatial coordinate sequence is then accumulated, analyzed, and statistically processed over a 10-minute observation period. Finally, the coordinate density distribution is used as the input for generating the heatmap.

Technical Scope: According to the academic requirements of this project, the proposed solution must operate independently without relying on pre-built pose estimation libraries such as MediaPipe. This constraint requires the research team to design and implement the entire processing pipeline from scratch, including object detection and foot position interpolation, based solely on the fundamental principles of Image Processing and Computer Vision.

CHAPTER II: METHODOLOGY

2.1 System Overview

The proposed system is designed to generate a heatmap representing the activity regions of a badminton player from a video recorded by a fixed camera. The entire processing pipeline is divided into three main stages: court detection and perspective correction, player detection and tracking, and heatmap generation based on the player's positional data.

First, the system receives the input video and processes it frame by frame. The badminton court is identified based on the color characteristics of the court surface. Subsequently, the court boundaries are extracted and used to construct a perspective transformation matrix. The output of this stage is a perspective-corrected image of the badminton court viewed from a bird's-eye perspective, allowing all positions on the court to be represented within a unified coordinate system.

Next, the system performs player detection and tracking within the identified court region. Once the player's position has been determined in each frame, the system estimates the location of the player's feet to approximate the contact point with the court surface. This position is then transformed into the court coordinate system using the perspective transformation matrix obtained in the previous stage.

Finally, the estimated foot positions are accumulated over time to generate the heatmap. Instead of representing each foot contact as a single point, the system models the contact region using an elliptical area to better approximate the actual footprint. The accumulated data are then smoothed using a Gaussian filter, followed by the application of a color map to visualize the activity density of the player across the court. The final output is a heatmap highlighting the areas where the player most frequently moves or remains throughout the observation period.

The overall workflow of the proposed system is illustrated in **Figure 2.1**.

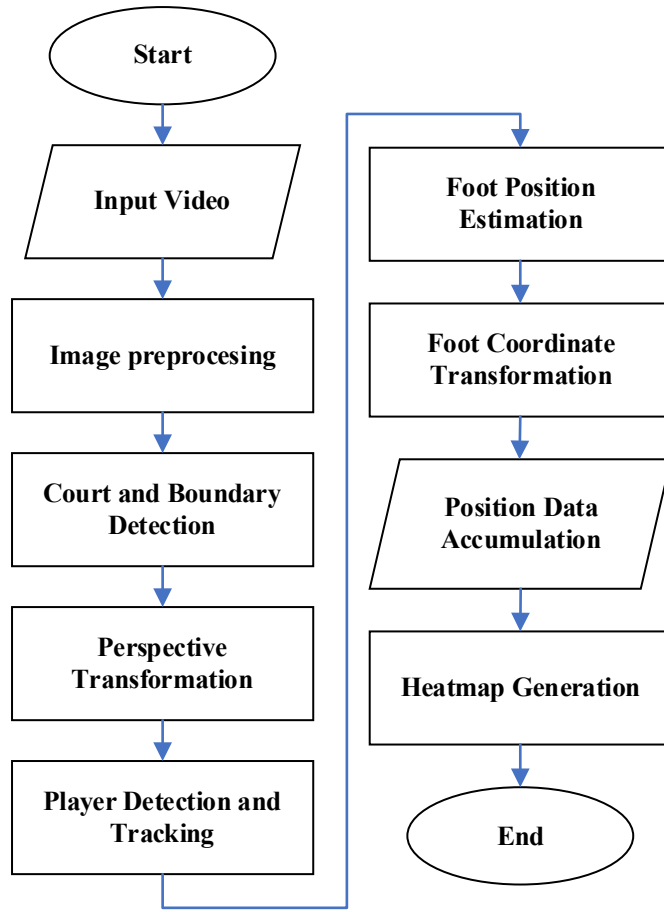


Figure 2.1: Overall Workflow of the Proposed System.

Specifically, the court detection and perspective correction results are used to transform the player's position into the standardized court coordinate system, while the tracked player position data serve as the basis for heatmap generation. Therefore, the accuracy of each processing stage directly affects the quality of the final output.

2.2 Court Detection and Perspective Correction

The first stage of the proposed system is court detection and perspective correction. The objective of this stage is to transform the badminton court into a bird's-eye view, thereby establishing a unified coordinate system for representing the player's position in the subsequent processing stages. If the original image coordinates were used directly, the resulting position estimates would be affected by perspective distortion caused by the camera being installed at an oblique angle relative to the court surface.

First, each video frame is converted from the BGR color space to the HSV color space to exploit the color characteristics of the badminton court. Compared with the BGR color space, the HSV model separates color information from brightness, making

it easier and more robust to define the color range corresponding to the court surface. After selecting an appropriate HSV threshold, color segmentation is performed to generate a binary image in which only the court region is preserved.

The binary image obtained after color segmentation still contains noise and small unwanted regions. Therefore, the morphological operations **Opening** and **Closing** are applied to remove noise while improving the continuity of the court region. After this preprocessing step, contour detection is performed, and the contour with the largest area is selected as the badminton court region.

Once the court region has been identified, the system proceeds to detect the court boundaries. First, the **Canny Edge Detection** algorithm is applied to highlight the edges of the court. The output of this step is an edge image containing the major court lines. Next, the **Hough Line Transform** is employed to detect straight lines from the edge image. Based on the detected line positions, the system selects the appropriate court boundaries and determines the reference points required for perspective transformation.

Using the extracted reference points, a perspective transformation matrix is constructed to map the court region from the camera view to a bird's-eye view. The resulting image is a perspective-corrected representation of the badminton court, in which every location is expressed within the same coordinate system. This unified coordinate system provides the foundation for transforming the player's position into court coordinates and generating the heatmap in the subsequent stages.



a)



b)



c)



d)



e)

Figure 2.2: (a) Original camera image; (b) HSV image after court region extraction; (c) Binary image after court boundary segmentation; (d) Hough Line detection result for the half-court; (e) Perspective-transformed image of the half-court.

2.3 Player Detection and Tracking

After completing court detection and perspective correction, the system proceeds to detect and track the badminton player in each frame of the input video. The objective of this stage is to accurately determine the player's position throughout the match while estimating the foot position, which serves as the basis for heatmap generation.

Since the court region has already been identified in the previous stage, image processing is restricted to the court area to reduce the influence of irrelevant objects outside the playing field. First, the image is converted to the HSV color space, and color thresholding is applied to generate a mask of the court region. The mask is then inverted to retain only the objects located on the court, among which the player is the primary target for detection.

The segmented image is further processed using morphological operations to remove noise and improve the continuity of the detected object regions. The system then extracts the contours of the remaining objects and performs candidate filtering based on the contour area, the aspect ratio of the bounding box, and the object's position within the image. These criteria effectively eliminate non-player objects and identify the most suitable candidate for tracker initialization.

After the initial player location has been determined, the system employs the CSRT (Discriminative Correlation Filter with Channel and Spatial Reliability) tracker to follow the player across consecutive video frames. The CSRT tracker provides relatively robust performance when the target undergoes pose variations, scale changes, or rapid movement, making it well suited for tracking badminton players during gameplay.

To reduce tracking drift, the system does not rely exclusively on the tracker. Instead, player detection is performed periodically or whenever the tracking result is considered unreliable. This re-detection mechanism updates the player's position and improves the overall robustness of the system, particularly when the player moves rapidly or becomes temporarily occluded.

Once the player's bounding box has been obtained, the system estimates the foot position. Rather than directly using the center or the bottom edge of the bounding box, the algorithm extracts the lower-body region corresponding to the player's feet and performs image segmentation to obtain the foot contour. The lowest points on the contour are then identified, and the center of this bottom region is regarded as the contact point between the player's feet and the court surface. This estimated foot position is subsequently used for heatmap generation in the next stage.

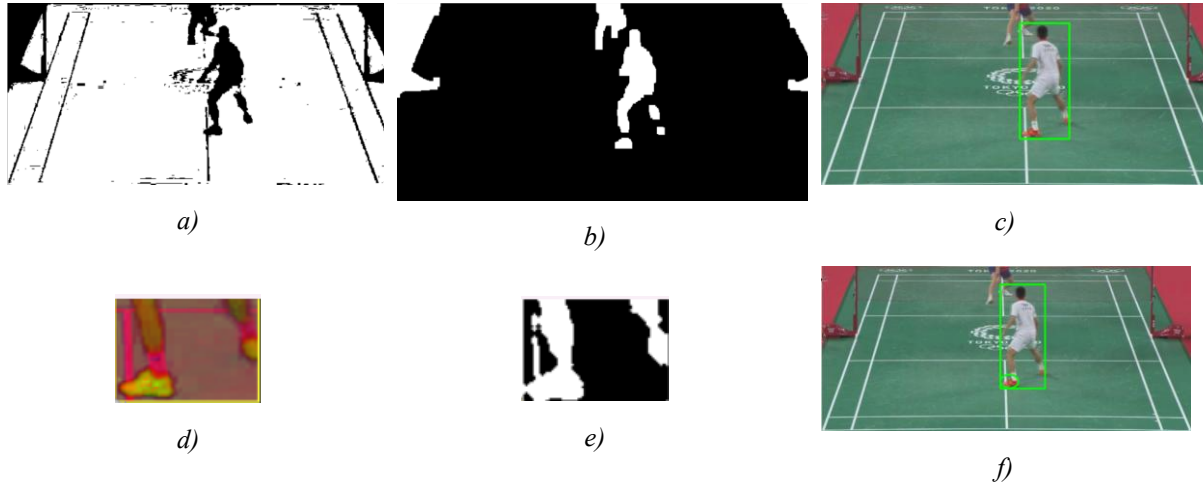


Figure 2.3: (a) Binary image of the court surface; (b) Inverted binary image with the court surface removed; (c) Player contour detection; (d) Cropped HSV image of the foot region; (e) Binary image of the foot region; (f) Foot contour detection.

2.4 Heatmap Generation

After estimating the player's foot position in each video frame, the system generates a heatmap to visualize the player's activity density on the badminton court. The heatmap highlights the areas where the player most frequently moves or remains during the observation period, thereby supporting tactical analysis and the evaluation of the player's movement patterns.

First, the estimated foot position is transformed from the original image coordinate system to the court coordinate system using the perspective transformation matrix obtained in the previous stage. As a result, all player positions are represented on the same reference plane and are no longer affected by the camera viewing angle.

Instead of representing each foot contact by a single point, the system models the contact area using an ellipse whose dimensions approximate the actual footprint on the court surface. This elliptical region is then projected onto the court coordinate system through the perspective transformation and accumulated into the heatmap matrix.

Compared with a single-point representation, this approach provides a more realistic description of the player's activity region and produces a smoother spatial density distribution.

During the accumulation process, a Gaussian Blur filter is applied to smooth the density distribution across neighboring regions, allowing the heatmap colors to transition continuously while reducing the appearance of isolated hotspots. In addition, the accumulated intensity of each pixel is limited using a clipping operation to prevent excessively large values from developing during long observation periods.

After the accumulation process is completed, the heatmap matrix is normalized and converted into a color image using the JET color map. Regions with low visitation frequency are represented by cool colors, such as blue and green, whereas regions with high activity density are displayed using warm colors, such as yellow and red. Finally, the generated heatmap is overlaid on the perspective-corrected court image to produce an intuitive visualization that enables users to easily observe and analyze the player's activity distribution on the badminton court.

CHAPTER III: RESULTS AND EVALUATION

3.1 Experimental Results

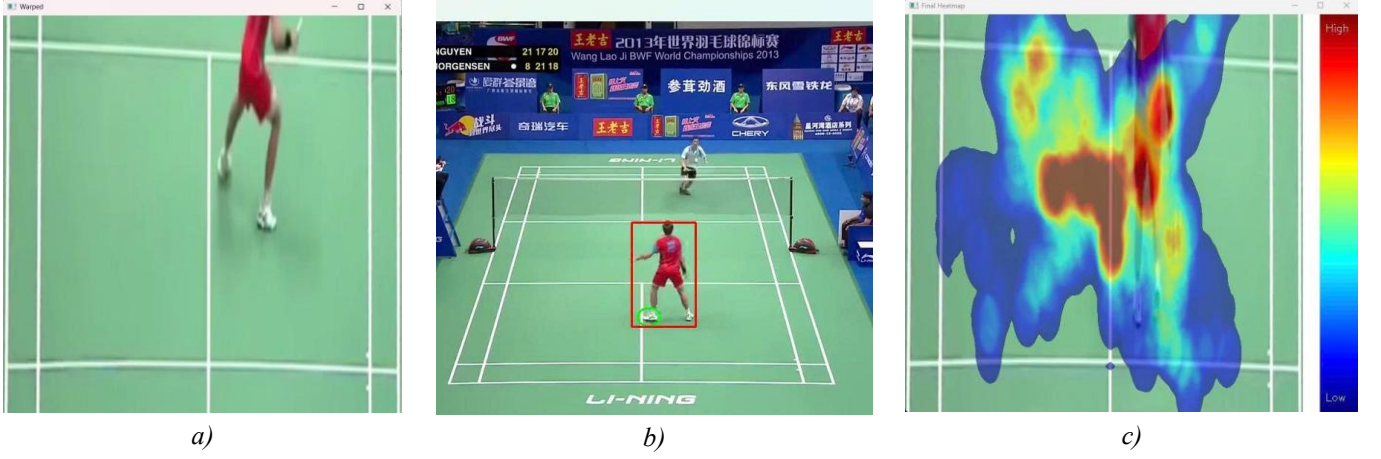


Figure 3.1: Heatmap generation process for Video 1. (a) Perspective-corrected court view (bird's-eye view); (b) Original video frame with player detection; (c) Generated heatmap showing the player's activity density.

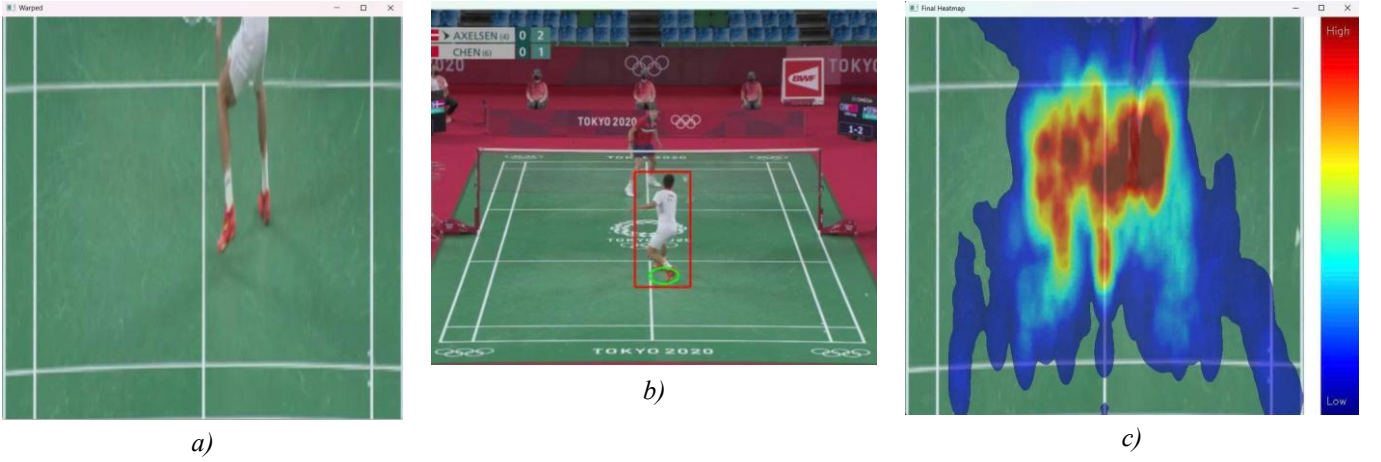


Figure 3.2: Heatmap generation process for Video 2. (a) Perspective-corrected court view (bird's-eye view); (b) Original video frame with player detection; (c) Generated heatmap showing the player's activity density.

3.2 Model Evaluation

❖ Accuracy and Performance Evaluation

Overall, the proposed processing pipeline operates reliably and fulfills the objectives of the project, achieving an overall accuracy ranging from approximately 80% to 90%. The performance of the system is evaluated for each major processing stage as follows:

- **Perspective Correction and Player Detection:** The perspective transformation successfully maps the target court region (half of the badminton court) to a bird's-eye view with high geometric accuracy. Meanwhile, the player detection algorithm

consistently establishes an accurate bounding box and successfully tracks the target player throughout the rally without losing the target.

- **Foot Position Estimation and Heatmap Generation:** The foot position estimation module, which represents the key reference point of the proposed system, achieves an accuracy of approximately 80%. Although some estimation errors remain due to the complex movement patterns of badminton players, the extracted positional data are sufficiently accurate for the subsequent interpolation and heatmap generation process. The resulting heatmap accurately reflects the player's movement distribution and activity density observed in the experimental video.

The experimental results demonstrate that the proposed approach not only overcomes the primary technical constraint of the project—namely, developing the system without relying on existing pose estimation libraries—but also satisfies all the academic objectives specified for the project.

❖ **Limitations of the Proposed Model**

Despite the encouraging results, the current system still exhibits several limitations arising from its reliance on traditional image processing techniques:

- **Limited Adaptability of Court Detection:**

The HSV color thresholding method used for background segmentation and court line detection is manually tuned for a specific environment. Consequently, the system's performance degrades under varying illumination conditions, such as strong sunlight, shadows, or changes in the court surface color.

- **Limitations of Player Detection:**

The player detection stage relies on color-mask inversion (`bitwise_not`) rather than feature-based object extraction. As a result, the system may fail to detect the player or produce fragmented contours when the player's clothing has a color similar to the court surface. Furthermore, external objects such as shadows, badminton rackets, or advertising boards may be incorrectly detected as players (false positives). The use of manually designed geometric constraints, including contour area and bounding box aspect ratio, also limits the model's ability to generalize across different viewing distances, player scales, and complex body poses such as lunging or jumping.

- **Limitations of the Tracking Algorithm:**

Although the CSRT tracker provides stable visual tracking, it does not explicitly recognize human objects. The high-speed movements typical of badminton frequently produce motion blur, causing the tracker to drift toward nearby objects. Moreover, the current tracker reinitialization mechanism relies on relatively simple thresholds based on changes in bounding box area, which may incorrectly interpret legitimate posture changes as tracking failures.

- **Errors in Foot Position Estimation:**

The foot localization algorithm extracts the lower portion of the bounding box and applies the same color-based segmentation strategy to estimate the foot contour. Under complex lighting conditions, strong shadows near the player's feet may be incorrectly identified as part of the foot, resulting in localization errors. Furthermore, when the player jumps or significantly changes body posture, the assumption that the feet remain on the court plane is violated. Consequently, the two-dimensional perspective transformation introduces positional errors into the projected foot coordinates, which may reduce the accuracy of the generated heatmap.